

Chapter 1 - Adaptive Measurement

Adaptive sampling changes the measurement interval according to signal variation. Rapidly changing signals are sampled more often, while stable signals are sampled less often.

The method reduces redundant measurements without ignoring short-lived changes. A valid controller therefore observes variation before changing the interval.

Chapter 2 - Stability Margin

The stability margin is calculated as $M = (\text{target} - \text{measured}) / \max(\text{abs}(\text{target}), \text{epsilon})$.

A positive margin means the measured value remains below the target. A negative margin means the measurement has crossed the target. Epsilon prevents division by zero.

Chapter 3 - Drift Response Table

The drift policy uses two operational bands.

Low drift | below 0.5 percent per hour | monitor normally.

High drift | above 2.0 percent per hour | recalibrate immediately.

Values between the two bands require a second observation before changing the calibration.

Chapter 4 - Feedback Diagram

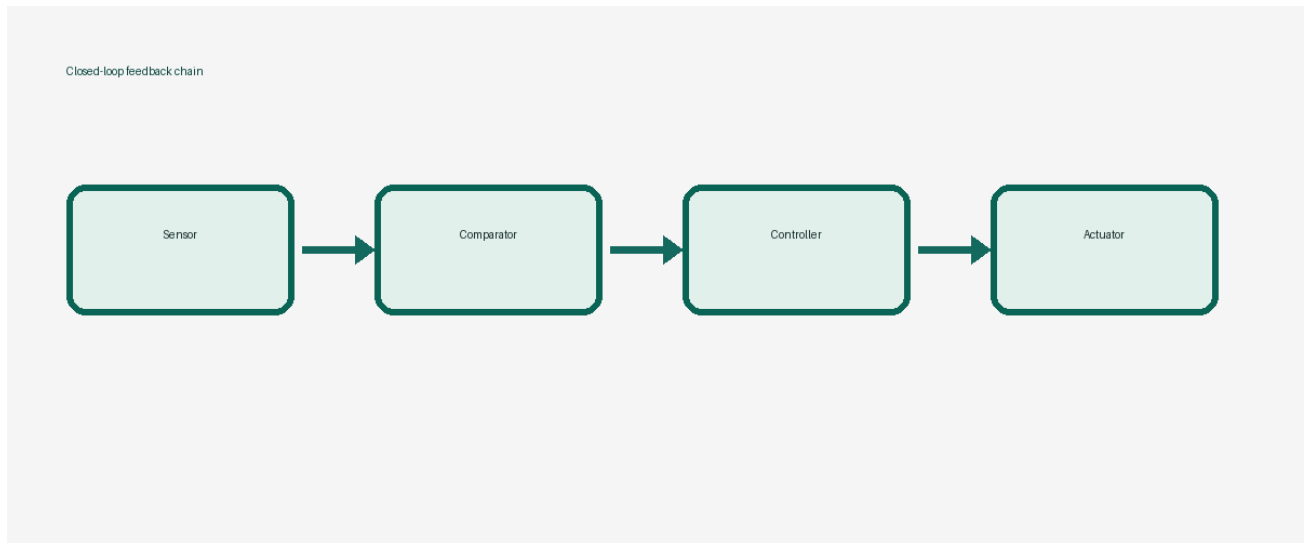


Figure 1 shows the feedback chain: sensor, comparator, controller, and actuator. The comparator measures error before the controller selects a correction.